

CHEM 405

Chemistry Principles for Engineers



Instructor:

Gordon Hunter, Ph.D.

Contact:

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Lecture: Online/asynchronous

Lab: Mondays, 6:10–9:00pm

Room: 574

Chemistry 405, *Chemistry Principles for Engineers*, is a one-semester course focused on those principles of chemistry relevant to the engineering disciplines. The goals of this course are three-fold. First and foremost, this course will be an introduction to chemistry—not just as a body of facts, but as an incredibly useful and interesting way of thinking about the world. This course will also build your knowledge of the fundamental ideas (principles) upon which chemical reasoning is built. These principles include the nature of matter and energy; the structure of the atom; how and why atoms assemble to form elements, compounds, and materials; understanding how molecular structure and composition determines the properties of materials; and the laws of chemical thermodynamics, its applications and consequences. Lastly, this course aims to build your skills in problem solving, analytical reasoning, and data analysis. At the end of the semester, you should begin “thinking like a chemist” by identifying chemistry’s central ideas; recognizing recurring relationships between these ideas; and applying these ideas and relationships to solve real problems, make correct inferences and deduce conclusions from data, and interpret whether conclusions are warranted based on given data. Chemistry 405 is required for all engineering majors. As such, this course will explore chemical phenomena and principles with emphasis on developing skills and knowledge applicable to engineering.

Course information

Required textbook:

OpenStax Chemistry: Atoms First, 2e

View online, free of charge:

<https://openstax.org/details/books/chemistry-atoms-first-2e>

Aktiv Chemistry online homework:

This is required, so please register for Aktiv right away. Instructions for registering are found in the Modules section on myCourses.

A note regarding your Aktiv grade:

Try to focus on understanding each question on Aktiv rather than combing the internet to find the correct answer. Your homework grades will be based on the rubric below:

Score earned in Aktiv	Score recorded in myCourses
90 – 100%	100%
80 – 89.99%	90%
70 – 79.99%	80%
60 – 69.99%	70%
< 60%	same as raw score

Please DO complete the whole assignment, though. They are good practice for the quizzes, and Aktiv offers feedback on your responses. End-of-chapter textbook questions are also good practice, but do not offer immediate feedback.

Office hours:

By appointment online using Zoom, but we can also talk in the lab during or after the lab activity.

Course website:

<https://mycourses.unh.edu>

All course materials will be posted on myCourses. Lecture slides will be available to download prior to each week's video lecture. I will also post additional practice problems and other useful material, so make sure you are able to access the course website.

A recommended study strategy for this course is: 1) read the assigned material in the text before interacting with the lecture material on myCourses; 2) view the lecture video(s) and take *your own* notes; and 3) as soon as possible after the lecture, begin reviewing learning objectives and working through associated Aktiv Chemistry exercises. When you encounter exercises that you cannot solve or ideas that are confusing, refer to the text, your notes, or your fellow students. Forming a study group to work through exercises and to talk about chemistry is an excellent way to master the subject. This recommended strategy, however, should not imply that chemistry is solely about working textbook exercises. *A solid conceptual understanding of chemistry will be the main goal of the course.*

Laboratory:

Handouts of the procedures and pre-lab questions will be provided in lab. Copies of these handouts will also be available to view on myCourses. You will also need to record and submit your data and observations for each experiment, but you will not need a dedicated laboratory notebook. A scientific calculator is also highly recommended for ease of calculations during the lab. Please refer to the separate lab syllabus (available on myCourses) for more information about the laboratory component of this course.

Calculators:

An inexpensive calculator will be required for exams (no cell phones permitted). It needs to be able to handle exponents and logarithms. A scientific/exponential notation function (may be labeled “E,” “EE,” or “EXP”) is very useful.

Grading:

Your course grade will be calculated as follows:

Aktiv Homework	30%
Midterm exam (online)	20%
Final exam (online)	20%
Lab activities	30%

- Homework and exams will be administered on Aktiv (see schedule for dates). Be sure to complete these tasks before each due date, as these dates are firm.
- There will be two exams this semester: (1) one midterm exam, given during the middle of October, and (2) one final exam, given during finals week in December. Details about each exam will be available on myCourses. Both exams will be online, in the Aktiv platform.
- Students must pass the lab section (with average 60% or better) to pass this course.

Final course grade:

Your final course letter grade will be assigned based on your course average, according to the cut-offs listed to the right. As the semester progresses, your assessment scores will be available on myCourses, so you can compare your course average to these cutoffs to get an idea of where your current grade is.

Grade	Range (%)
A	≥93.33 to 100
A-	≥89.95 to <93.33
B+	≥86.67 to <89.95
B	≥83.33 to <86.67
B-	≥79.95 to <83.33
C+	≥76.67 to <79.95
C	≥73.33 to <76.67
C-	≥69.95 to <73.33
D+	≥66.67 to <69.95
D	≥63.33 to <66.67
D-	≥59.95 to <63.33
F	0 to <59.95

See next page for course schedule.

Date/Week	Topics	Reading in <i>OpenStax</i>	Laboratory	Assessments
Aug 25	Chemistry: the nature of matter and energy	1.1-1.6	No lab	
	Connecting the macroscopic to the microscopic	2.1-2.4		
Sept 1	Electrons and <i>electron configurations</i> , predicting behavior and properties using electron configurations	3.1-3.7	No lab – holiday	
Sept 8	The structure of molecular and ionic solids, and their names	4.1-4.3	Lab safety and procedures	
Sept 15	A chemically useful model: <i>Lewis structures</i>	4.4	Lab #1: The atomic spectrum of hydrogen	
	Shapes of molecules; polarity of covalent bonds and molecules	4.6		
Sept 22	Attractions between molecules	10.1	Lab #2: Resolution of matter I: Paper chromatography	
Sept 29	Chemical reactions and stoichiometry	6.1, 7.1-7.4	Lab #3: The geometrical structure of molecules	
Oct 6	Catch up / study for midterm exam		Lab #4: Resolution of matter II: Fractional crystallization	
Oct 14	Organic molecules	21.1	Online exam – no lab	Midterm exam
Oct 20	Energy and the first law of thermodynamics; measuring and using enthalpy	9.1-9.3	Lab #5: Determination of an unknown chemical formula	
Oct 27	Entropy and the second law of thermodynamics	12.1-12.3	Lab #6: Molar mass from freezing point depression	
	A chemically useful model: <i>Gibbs energy</i>	12.4		
Nov 3	Rates of reactions and rate laws	6.3, 17.1-17.3, 17.6	Lab #7: Heat effects and calorimetry	
	Engineering connection: <i>catalysis</i>	17.7		
Nov 10	Chemical equilibrium	13.1-13.2	Lab #8: Spectroscopy – using light as a probe	
Nov 17	Chemically useful models: <i>equilibrium constants and Le Châtelier's Principle</i>	13.2-13.3	TBD	
Nov 24	Practical equilibria	14.1-14.3, 14.6-14.7	TBD	
Dec 1	Review for final exam		TBD	
Dec 8			Online exam – no lab	Final exam

Important Academic Policies

Academic Alerts

The University is invested in your academic success. If a faculty member is concerned about your academic behavior or performance, they may submit an academic alert. Academic alerts are not punitive. The goal is to provide you with support and resources to support your success. They act as an important check-in point and, if you receive an academic alert, you will receive an email to your UNH email address. It is strongly recommended that you meet with a professional advisor and connect with your instructor to discuss the reason for the alert.

Accommodations for Disability

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Student Accessibility Services (SAS) office. The Student Accessibility Coordinator at UNHM is Jenessa Zurek. Please reach out to the SAS office via email at jenessa.zurek@unh.edu for registration information and disability related questions. Jenessa Zurek is available through phone and email Mondays and Wednesdays from 9am-2pm.

COVID-specific Policies and Guidance

Health and Safety in Face-to-Face Classes

As faculty, TAs, and course instructors, we need your help to promote our [#unhtogether](#) COVID response culture. We all have a responsibility during this COVID-19 pandemic to protect our own health and the health of friends and fellow community members. Violations of the COVID protocols by even a single individual can cause significant disruptions or discontinuation of in-person academic activities. Any student creating such disruptions undermines the opportunity for others to learn and engage with the UNH community, and as such, is in serious violation of the UNH [Student Rights, Rules & Responsibilities](#).

In alignment with our [#unhtogether](#) commitment to the ongoing health and safety of our community during the COVID-19 pandemic, there are several changes in classroom expectations.

All students are required to wear masks in class and campus buildings unless a medical exception is made through an accommodation process. It is your responsibility to obtain a mask before coming to class. For information on proper use of masks, acceptable mask types, and other PPE and social distancing guidelines visit <https://www.unh.edu/coronavirus>. Students wishing to request a medical accommodation should contact the [Student Accessibility Services](#). Failure to comply with PPE and social distancing classroom protocols is a violation of the [Student Rights, Rules & Responsibilities](#). If you refuse to comply, you will be asked to leave class immediately and you may also be reported to the Office of the Dean of Students and your associate dean.

1) Your instructor or TA may be wearing a face shield without a mask during instruction and only while maintaining at least a 6-foot distance from any student.

2) Prior to class, please wait outside the building, weather permitting, or in the hall or common area, observing social distancing and leaving plenty of room for the prior class to exit the room and building. Wipes are available near the room entrance. Obey entrance and egress signage and any additional faculty directions on entering or leaving the classroom.

3) Each classroom entrance is equipped with hand sanitizer and surface wipes.

- Use hand sanitizer as you deem appropriate.
- Wipe down your personal space prior to class and throw the used wipes away on the way out of class or take them with you.

4) Sit only in marked seats. Classes and laboratories were restructured to minimize or eliminate contact between individuals of less than 6 feet.

5) Contact tracing is mandated by the State of New Hampshire. The following practices will facilitate effective contact tracing implementation should the need arise:

- Students should sit in the same seat for each class period to minimize potential contacts.
- We ask students to know the names of each of the people sitting closest to them — their nearest neighbors.

6) UNH has developed "Wildcat Pass," a web and mobile app to help each of us keep track of the requirements for being in compliance with necessary testing, isolation and quarantine rules that will help to keep our community healthy.

- Log into your Wildcat Pass each day.
- Be prepared to show your mobile device or a computer printout of your daily Wildcat Pass if asked by a university representative.

Changes to Course Modalities

1) This class is beginning the semester operating in a *yellow (restricted)* mode of operation ([see https://www.unh.edu/coronavirus/operation](https://www.unh.edu/coronavirus/operation)). This means we are meeting in person for lab activities. If your health and safety requires shifting to an Orange or Red mode of university operation, the modality and schedule of this course may change.

2) All in-person class meetings will end on Friday, Nov. 20, during the Fall 2020 semester and will move to a remote modality starting Monday, Nov. 23. The type of remote learning we will use in this course is *asynchronous* activities available on Canvas.

3) Your classroom is equipped with technology that will provide remote access to class instruction. Because of the need to accommodate potential isolation and quarantine due to the COVID pandemic,

lectures or other class meetings for this course may be recorded by the university using UNH media platforms. Such recordings may be available for educational use by students enrolled in the class (including both for instruction and as a review tool), the course instructor(s), and other university officials who support course instruction. Your voice or image may be captured on the recordings, and by enrolling in this course you are consenting to such recording for these purposes. The university and Zoom have FERPA-compliant agreements in place to protect the security and privacy of UNH Zoom accounts. You may not share recordings outside of this course. Doing so may result in disciplinary action.

4) Students must learn how to access this course in all possible formats.

- Ensure that you have all necessary technology to participate in this course remotely.
- Consult with the instructor and/or with [UNH IT for Students](#) with questions.

Attendance (including rotational) for in-person modalities

5) Students are expected to adhere to the attendance policies specific to this course as outlined in the syllabus, as usual. Do not attend class if you have any symptoms of illness or if your daily Wildcat Pass does not show that you are cleared to participate in classes and other campus activities. Inform the instructor or TA, in advance if possible, that you will be absent from a scheduled in-person class. It is ultimately your responsibility to keep up with all course expectations. When appropriate, accommodations will be made.

If this class either begins with, or is forced by conditions to adopt, a rotational attendance component, you must only attend in-person on your assigned day(s) and may not switch days with other students or make other modifications to faculty instructions for in-person attendance.

Exam Scheduling and Remote Proctoring

6) This class will use myCourses and lockdown browser to administer and proctor exams. Within the first weeks of the semester, we will run a sample test using the required software. You will be asked to agree to the use of lockdown browser, webcam, recording of video, audio, etc. during the sample test. This software will be used for the remainder of the semester for quizzes/tests. If you have any problems with the use of myCourses and lockdown browser, contact your professor to discuss reasonable alternatives. It is my goal to create a learning experience that is as accessible as possible. If you anticipate any issues related to the testing requirements of this course or need accommodations, please either discuss them directly with me or in conjunction with the [Student Accessibility Services Office](#) within the first week of classes to explore alternative options.

Non COVID-specific Guidance

Netiquette Guidelines

Netiquette is the socially and professionally acceptable way to communicate on the internet. We are all expected to abide by these guidelines of "netiquette" when using online communication tools with

your classmates and instructor. Guidelines can be found on the Student Resource Hub and the Faculty Resource Hub, and in this [knowledge base netiquette guidelines article](#).

Affirming Credit Hour Compliance

This syllabus reflects the federal definition of a credit hour, which entails a minimum 3 hours of engaged time per week per credit over a 14-week semester. Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours. For more information, please see: <https://www.neche.org/wp-content/uploads/2019/01/Affirmation-of-Compliance.pdf>

Addressing Disabilities

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate his/her disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please see more information here - <https://manchester.unh.edu/academics/academic-services/student-accessibility-services> Please follow-up with me as soon as possible to ensure timely implementation of the identified accommodations in the letter.

Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS, but are under no obligation to provide retroactive accommodations.

Addressing the Needs of Students of all Faiths

In the event that a student needs accommodation for a religious or cultural holiday/observance, that student is encouraged to make that request as early in the semester as possible.

Title IX Confidentiality and Mandatory Reporting

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's Title IX Coordinator ([Donna Marie Sorrentino](#), (603) 862-2930/1527 TTY) any incidents of sexual violence and harassment shared by students. If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can [find a list of resources here](#). For more information about what happens when you report, how the university considers your requests for confidentiality once a report is made to the Title IX Coordinator, your rights and report options at UNH, including anonymous report options, please [visit the student reporting optionspage](#).

Referral Resources for Students

Library: The UNH Manchester librarians are available to assist you with your research. You can contact a librarian by calling 603-641-4173 or by emailing unhm.library@unh.edu.

The following online resources provide information about library resources and services:

UNH Manchester Library webpage: <https://manchester.unh.edu/library>

Online Research Guides: <https://libraryguides.unh.edu/index.php?b=s>

Access Library Resources Remotely: <https://libraryguides.unh.edu/remoteaccess>

Reserve a study room for Zoom classes: <https://libraryguides.unh.edu/remoteaccess/studyrooms>

Center for Academic Enrichment: Provides academic support services, including individual tutoring, for all students. Make an appointment through the CAE My Courses site on your Canvas dashboard. Phone: (603) 641-4113. Email: unhm.cae@unh.edu

Food Pantry: The campus food pantry has pre-packed bags of non-perishable food items for students in room 437. If you have any questions, contact Kattarina.Biss@unh.edu

Sexual Harassment and Rape Prevention Program (SHARPP): Provides free and confidential advocacy and direct services to survivors. (<https://www.unh.edu/sharpp>.)

QPR is a training program in mental health awareness and suicide prevention training offered by trained facilitators and members of the UNH Manchester community. Please contact Lisa Enright at lisa.enright@unh.edu should your department or program want to schedule a training session.

Behavior Intervention Team (BIT): This team provides assistance to the UNH Manchester community when there is a need to activate a systematic, coordinated response to students who may be in crisis or whose mental, emotional or psychological health condition may substantially disrupt or directly threaten the safety of the learning environment. The BIT receives reports from security, students, faculty, administrators, and others regarding students of concern, develops and implements appropriate interventions, assists students in accessing appropriate resources and recommends appropriate actions to the Dean of Students when needed. The UNH Manchester BIT is a collaborative interdisciplinary advisory team comprised of the following members:

Lisa Enright, Director of Student Engagement, Chair

Dan Reagan, Associate Dean of Academic Affairs

Stephanie Kirylych, Director of Academic Advising

Bruce Azotea, UNH Security in direct collaboration with UNH Police

More information can be provided by emailing the UNH Manchester BIT at UNHM.BIT@unh.edu. A member of BIT will follow up with you via email to confirm receipt of your concerns and will determine the proper referral procedure in any given case in consultation with BIT members. Please do not include the student's name in the email you send to her. Any BIT member can also be a helpful resource to you. BIT will work as close partners with faculty to provide a caring, holistic approach to promote student safety and well-being on the Manchester campus.